University of Westminster
School of Computer Science and Engineering

Subject to external examiner approval

5BUIS017C, "BIS Design and Architecture" (2025/26)	
Module leader	Janice Abeykoon
Unit	Group Coursework
Weighting:	60%s
Qualifying mark	30%
Learning Outcomes Covered in this Assignment:	**LO2** Understand and be able to apply different architectural styles and develop a design model for a given medium-scale business problem; **LO5** Engage confidently in academic and professional discussions, to reflect on their team-work experience, and to justify their business solutions and suggestions.
Handed Out:	**Week 3**
Due Date	Thursday, 04 December 2025, 13:00 (Week 11) Meeting Record Files (related to task 5): • Meeting 1: 30/10 • Meeting 2: 20/11 (about "patterns") • Meeting 3: 04/12
Expected deliverables	Electronic files
Method of Submission:	Upload your answers to Blackboard. More specifically – submission Instructions: • **Group Submission** (Tasks 1, 2, 4 & 5): One common file per group, including answers to Tasks 1, 2, 4, and 5. Only one group member should upload the file. Deadline: Thursday, 04 December 2025, 13:00 (Week 11) Submission: Group submission link on Blackboard • **Individual Submission** (Task 3): Each student must submit their own answer to Task 3 individually. This is a separate submission from the group file. Deadline: Thursday, 04 December 2025, 13:00 (Week 11) – same deadline as the group submission. Submission: Individual submission link on Blackboard • **Meeting Record Files** (related to task 5): Submit one file per meeting per group using the provided template. Only one group member should upload each file. Deadlines: Meeting 1: 30/10 Meeting 2: 20/11 (about "patterns") Meeting 3: 04/12 Submission: Separate submission links on Blackboard

	Electronic submission via Blackboard. All the submission links will be under the "Assessment: Information & Submission Links" section on BB.
Type of Feedback and Due Date:	• Verbal feedback during the tutorials • Written feedback and marks 15 working days (3 weeks) after the deadline. **All marks will remain provisional until formally agreed by an Assessment Board.**
BCS Criteria meeting in this assignment	2.1.1 Knowledge and understanding of facts, concepts, principles & theories 2.1.2 Use of such knowledge in modeling and design 2.1.3 Problem solving strategies 2.1.4 Analyse if/how a system meets current and future requirements 2.2.4 Deploy tools effectively 2.3.1 Work as a member of a development team 2.3.2 Development of general transferable skills 3.1.3 Knowledge of systems architecture 3.2.2 Defining problems, managing design process and evaluating outcomes 3.2.3 System Design 4.1.1 Knowledge and understanding of scientific and engineering principles 4.1.2 Knowledge and understanding of mathematical and statistical principles 4.2.2 Defining problems, managing design process and evaluating outcomes

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, **10 marks** will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 40 – 49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

The above applies to *each piece of work* you are required to submit for the module, including both group and individual submissions.

It is recognized that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Registry Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form unless you are submitting a self-certified claim. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regardingUniversity Assessment Regulations, please refer to the following website: **http://www.westminster.ac.uk/study/current-students/resources/academic-regulations**

Coursework Description

The coursework is about an online company. After a short description of the environment, your tasks for the assignment are listed. This is a group coursework with one individual task.

Scenario: "Foodiverse" – A Global Recipe Sharing Platform

Background and user types

"Foodiverse" is an online platform (available on both web and mobile devices) where users can browse and share recipes from different cultures and cuisines. It is a free platform, and users can either browse anonymously or create a personal account to access more features.

According to the level of interaction and engagement users have with the platform, they may hold one of the following roles:

Guest: A guest has read-only access. They can browse publicly available recipes and use basic filters but cannot save information, upload content, or interact with other users. Guests do not have a personal account in the platform.

Registered User: A registered user must create an account and a personal profile. Registered users can save recipes and dietary preferences to their profile, write comments and rate recipes, and create lists of favourite recipes and grocery ingredients.

Contributor: A contributor is a registered user who is also allowed to upload original content (recipes). A registered user becomes a contributor after maintaining consistent engagement with the platform for at least one month through active use, including commenting, saving, and interacting with content. Contributors can upload and edit their own recipes, include multimedia content (such as images, videos, and tips), and track engagement metrics on their content.

Distinguished Member: A distinguished member is a contributor recognized for high engagement and quality contributions. A contributor is upgraded to this role automatically when they have uploaded more than 15 recipes and have at least one recipe rated 4 stars or higher by other users. Distinguished members have access to exclusive features such as publishing long-form articles (e.g. cooking tips, regional cuisine guides) and advising or mentoring new contributors through comments or suggestions.

Key Features

Recipe Management

- Registered users can upload new recipes. A receipt must include (compulsory): ingredients list, preparation and cooking time, level of difficulty and regional information.
- Recipes can include optional **enhancements**, like:
  - Images
  - Nutritional breakdown
  - Video walkthrough
  - "Chef Tips" section

Filtering and Search

- Users can filter recipes by:
  - Specific ingredients (e.g., vegan, nut-free, keto)
  - Cuisine (e.g., Indian, Italian)
  - Preparation and cook time
  - Difficulty level
- Filtering logic varies based on user's preferences and goals.

Grocery List Generator and Grocery List Optimization Strategy
Based on saved recipes, users can generate a grocery list. When generating a grocery list from multiple saved recipes, users can choose how the list is organized or optimized, such as:

- By category: Grouped into produce, dairy, etc.
- By store section: For easier in-store navigation.
- By recipe: Items grouped under each recipe.

Registered User Profiles with Custom Features
When users create an account, they must create a basic profile with name, avatar, and at least two preferences. The profile can be enhanced with the following features:

- Achievements (e.g., "Uploaded 10 recipes" badge)
- Social Media Integration
- Meal Plans (create and publish daily or weekly meal plans)
- Personal Blog Section (write cooking-related articles)

These are optional features layered on top of a basic profile and can vary from user to user.

Distinguished Member Profiles with Custom Features
Contributors who upload more than 15 recipes and receive at least one 4-star rating are automatically upgraded to Distinguished Member status. A Distinguished Member has a basic profile that includes a name, avatar, at least two preferences, 15 published recipes, and a 4-star rating. This profile can be enhanced with the following features:

- Long-Form Articles: Publish detailed content such as cooking tips, regional cuisine guides, or personal stories.
- Mentorship Tools: Support new contributors through feedback and suggestions. May include badges like "Mentor" or "Top Advisor."
- Expertise Tags: Display areas of specialization (e.g., "Vegan Baking," "Mediterranean Cuisine") on your profile and recipes.
- Featured Section: Highlight selected recipes or articles and showcase recognition badges such as "Editor's Pick."

These are optional features layered on top of a basic profile and can vary from user to user.

App Settings (Global State)
Users can customize global settings such as language, measurement units, and theme. The app supports multiple languages, including English, French, and Spanish. Users can also choose between metric and imperial units for measurements. Additionally, the app's appearance can be adjusted by selecting a preferred theme.

Your tasks

All group members are expected to contribute to each of the following tasks. Only Task 4 is an individual task; however, group collaboration is still required to decide how the work will be divided among members.

Question on class diagram – 20 marks

Task 1: Produce a detailed design class with all the semantics necessary for the implementation, including all the boundary, control and entity classes (if there are any) as well as any other notation(s) required (Diagram:10 marks). Provide a paragraph (no more than 300 words) that explains your diagram (5 marks).

Question on state machine diagram – 20 marks

Task 2: Based on the scenario described above, users may hold different roles, and their states can change depending on their level of interaction and engagement with the platform. Roles such as "guest", "registered user", and others may be assigned. Draw a detailed state machine diagram for an object of the class "user". (15 marks). Make sure that in your diagram you include all the possible states, all relevant transitions, conditions or triggers as well as any other detail that is important. Provide a paragraph (no more than 300 words) that explains your diagram (5 marks).

Question on patterns – 25 marks. Individual task – must be submitted by each student separately using a specific submission link on Blackboard.

Task 3: In the description above, there are several cases where design patterns could be applied. You must work as a group to identify these cases. Then, each student must select a different case and, working individually, identify the design pattern that applies and create the relevant diagram.

Each student must submit:

a) the diagram of the pattern that describes their selected case (10 marks), and
b) a report—no more than 300 words—that explains why this pattern is appropriate for the case, what it does, and the structure of the diagram (15 marks).

There are a total of six cases, each involving one of the design patterns discussed in lectures and tutorials (Decorator, Strategy, and Singleton). Each student must upload their submission for Task 3 individually via the designated Blackboard submission link.

Question on Graphical User Interface design – 20 marks

Task 4. The diagrams presented above (1 and 2) are related to the structural and functional design of the online system. This design must now be "translated" into a graphical user interface. For this task, you are required to write a report of no more than 500 words describing the graphical design of the web page where a registered user uploads a new recipe, including all necessary information and optional enhancements.

In your report, you must include information on the following:

a) Content: Describe the content that needs to be presented on the screen (4 marks).
b) Colour Scheme: Explain your approach to choosing a colour scheme, including how you will ensure the combination is visually appealing and accessible. Be sure to mention a tool you could use to check and confirm a suitable colour palette (4 marks).
c) Design Standards and Conventions: Demonstrate awareness of design standards and conventions commonly used in web interface design. Select two standards or conventions, and for each, explain its purpose and how it would be implemented in your design (8 marks – 4 marks each).

Finally, your report must include references to support your design decisions. The references should be drawn from academic literature. Proper referencing is required. At least four references are expected (4 marks).

Records on group collaboration – proof of group work (15 marks)

Task 5: Throughout the entire coursework period, from assignment (week 3) to final deadline (week 11), your group is expected to collaborate consistently — not just in the final few days. You should arrange and hold multiple meetings, either in person or online, using any platform or communication tool that suits your group.

You must provide evidence of your group work ***throughout the coursework period***. This includes:

Organizing regular meetings

Recording the progress and discussions of each meeting

Demonstrating equal and active participation from all group members

As part of Task 5, you are required to submit on Blackboard, **three (3)** meeting records showing when your group met and what was discussed.

Each file must reflect one meeting

You must use the **provided template at the end of this file** (page 10) to record each meeting; it is also uploaded on BB as a separate file.

Each file must be uploaded separately, on the following dates:

Meeting Record	**Deadline**
Meeting 1	30/10
Meeting 2	20/11 (discussion on patterns for task 4)
Meeting 3	04/12

Note1: The second meeting for which you submit a record is expected to be about "patterns" and to show how you decided to choose patterns (individual task 4).

Note 2: It is advisable to keep records for all your meetings, so you can easily track your group's progress. However, you are only required to submit **three** records, according to the schedule above.

All group members are expected to attend and actively participate in each meeting. To monitor participation, you must accurately complete the meeting record for each session using the template provided.

Penalties for late submission of the files for group meetings:

If the files are uploaded after the intermediate deadline, a 50%-mark reduction will apply for all members of the group. If one or more files are uploaded after the **final deadline (04/12/2025)**, no marks will be awarded.

Penalties for non-participation:

- A member who misses one meeting, gets 10% penalty on group work (Tasks 1, 2, 4, and 5)
- A member who misses two meetings, gets 20% penalty on group work (Tasks 1, 2, 4, and 5)
- A member who misses all meetings, gets 50% penalty on group work (Tasks 1, 2, 4, and 5)

Apart from the records you submit as part of Task 5, you should write a brief paragraph (no more than 200 words) outlining the total number of meetings you held (at least three) and evaluating the success of your collaboration. Task 5 will be assessed based on the quality, accuracy, and level of detail in your meeting records. If the records have been uploaded but no

paragraph is submitted as part of Task 5, the group will lose 5 marks.

VERY IMPORTANT NOTE: All diagrams must be produced using appropriate software.You could use the software I suggested in the file "5BUIS017W Software Required.pdf" on Blackboard or **any other** software you prefer. Hand-drawn diagrams are **not** allowed, and they will get **NO** marks.

Viva

Viva is not a requirement for this coursework. Only in exceptional cases (e.g. if there are complains for the team work) a viva might be organized during tutorial week 12.

What you must submit – Submission Instructions

- **Group Submission** (Tasks 1, 2, 4 & 5):
  One common file per group, including answers to Tasks 1, 2, 4, and 5.
  Only one group member should upload the file.
  Deadline: Thursday, 04 December 2025, 13:00 (Week 11)
  Submission: Group submission link on Blackboard
- **Individual Submission** (Task 3):
  Each student must submit their own answer to Task 3 individually.
  This is a separate submission from the group file.
  Deadline: Thursday, 04 December 2025, 13:00 (Week 11) – same deadline as the group submission.
  Submission: Individual submission link on Blackboard
- **Meeting Record Files** (related to task 5):
  Submit one file per meeting per group using the provided template.
  Only one group member should upload each file.
  Deadlines:
  Meeting 1: 30/10
  Meeting 2: 20/11 (about "patterns")
  Meeting 3: 04/12
  Submission: Separate submission links on Blackboard
  If the files are uploaded after the intermediate deadline, a 50%-mark reduction will apply. If one or more files are uploaded **after the final deadline** (04/12/2025), no marks will be awarded.

Electronic submission via Blackboard. All the submission links will be under the "Assessment: Information & Submission Links" section on BB.

Marking Scheme

Criteria	Mark per component	Mark given	Comments
Task 1: detailed class diagram Max 3 marks for correct classes (1 mark if half of the required classes are included) Max 3 marks for correct set of operation in the classes Max 3 marks for correct set of attributes in the classes Max 3 marks for correct associations 3 marks for correct use of symbols 5 marks for the explanation (report)	**20**		
Task 2: State machine diagram Correct states: 5 marks Events: 5 marks Transactions and symbols: 5 marks 5 marks for the explanation (report)	**20**		
Task 3. Patterns – individual task Detailed pattern diagram: 10 marks Report: Explain why the specific pattern is appropriate for the class selected: 5 marks. Give a general description of the pattern and explain what it does and in which cases we use it: 5 marks Explain in detail the diagram you created: 5 marks *Each student must use a different case. There are enough cases (6 in total) where a pattern can be applied.*	**25**		

Task 4: Graphical User Interface design Content: the information that should be presented on the screen: 4 marks. Colors: explain how you will decide the color combination and refer to a tool you could use to confirm the good combination of colors: 4 marks Design standards and conventions: select any two (2) of the standards and/or the conventions we use in web graphical design, present them, and explain how you could implement them in your interface: 8 marks – 4 marks each. References: 4	**20**		
Task 5: Report on group collaboration – proof of group work As part of Task 5, you are required to submit on Blackboard, **three (3)** meeting records showing when your group met and what was discussed and a brief paragraph (up to 200 words). Read carefully pages 6 and 7 above for details and specific deadlines.	**15**		
TOTAL	**100**		

Please notice that your final mark depends on the correctness, quality, and completeness of your answers.

Group Meeting Minutes Template

Module Title:
Group Name/Number:
Meeting Number (e.g. 1st, 2nd):
Date:
Time:
Location / Platform: (e.g., in person / Zoom / Teams)
Attendees: (List all group members present)
Absent Members: (Optional)

Topics to be discussed

(List topics to be discussed, either planned before the meeting or raised during it)

1.
2.
3.

Discussion Summary

Topic Discussed	Key Points / Decisions Made	Person Responsible (if any)	Person raised the topic
		e.g. All members	

Action Items

Task	Assigned To	Status	Due Date
	e.g. All members	e.g. completed	

Next Steps / Reminders

- (e.g., All members to upload their sections to the shared drive before the next meeting)
-

Next Meeting

Date & Time:
Location / Platform:
Proposed Agenda Items:

1.
2.

Minute Taker: (Name of the person who recorded the minutes)